

ActInf GuestStream #006.1 -- Anna Ciaunica

GuestStream | 1:22:49

hello

let me just wait one sorry just one

second

so you introduced me first right yep

just wanted to make sure that it started

and it looks like

we have so hello and welcome

everyone to active lab this

is actin flap guest stream number 6.1

and we're really happy to be here with

dr anna cia

so anna thanks so much for joining we

really appreciate it

and we're looking forward to this talk

please just take it away and if anyone

in the

uh live chat wants to ask a question i

will relay those questions to you

at the end of this talk so thanks again

for coming on the stream

and we're looking forward to what you

have to share

thank you so much danielle and thank you

so much for having me

um today i'm going to present a paper

that we worked

on the past past year and just came out

um but i'm going to start very briefly

with some of my background as a

researcher

so i'm trained as an armchair

philosopher so i did um in analytics

philosophy so i did my thesis on

physicalism in qualia

you know the famous mind body problem so

how exactly so some physical stuff
connects with
you know mental stuff and gets you know
the uh
gives rise to the subjective experience
of our conscious experiences like the
heart
problem of consciousness um
and i was obsessed as any respectable
philosopher i reality philosophy with
you know the whole problem of
consciousness and whether there is
something
special about the subjectivity of our
conscious
experiences right and because i'm coming
from burgundy
i spent a lot of time trying to you know
say that there is something about our
there is something special about our
subjective conscious experience because
there is something what is this like for
instance to taste
a particular wine and coming from
burgundy i had a lot of
experience and natural example with that
but more recently i
decided to get off the armchair of
philosophy
and i'm recently started to collaborate
with people from
quantum science i actually also have a
master in cognitive science
in development of psychology and uh
i'm recently collaborated with the
institute of community of neuroscience
as ucl in london
and my job there is to ask annoying

questions to

uh scientists like what do you mean by
perception what do you mean by social
right

um just to give us some examples so uh
one of the annoying questions i really i
like to ask

is um you know whenever i go to a
conference

is uh i ask people to look around them
in the audience and realize that
actually there are no infants in the
audience

uh just adults and uh
i interpolate them on that and they look
at me say

you know why i don't see the point why
is this important

well it is important because i mean when
we start investigating mind and bodies
we already have another centric
um theoretical lens so this means that
the way we look at the phenomena
is from a certain standpoint and the
standard

point that you're using inverted
um inescapably is an adult one
right so this means that we might have a
sort of like

implicit bias towards certain phenomena
a certain way of

understanding phenomena that is somehow
you know

um adult centric and uh we need to
be careful if we want to understand her
mind and

uh how mind and brain work in the first
place so

um given that my initial uh
you know entrance point into the science
and philosophy was like
uh perceptual awareness like the
fundamental point of contact
of a subject with the reality of the
objects out there in the world
so um this is like the fundamental way
of actually structuring an experience
so you have this axis you have an
experience of subject with relations
object over there
and typically for instance this is how
self-consciousness
is understood like uh perceptual
awareness when
one perceives oneself as self-distinct
from the world and others
and how is that linked to
self-recognition in the mirror for
instance right
so for instance like is the famous rouge
test if you recognize yourself in the
mirrors as a toddler
and you see that the person in the
mirrors is you then you you know take
your
post-it or rouge from your uh forehead
and that's supposed to
give you a sign that you are basically
self-conscious
and self-aware but what i want to say is
that
maybe by at an um and dorsey that
adult-centric perspective we
tacitly also endorse a sort of like
visual spatial perspective on what
perceptual awareness is from the

um from the very beginning so
um while i i was doing my first post
talk
on in philosophy on
uh the sensory motor basis of the self
i also started to do a mastering
cognitive science and
development of psychology and i started
to actually address
the question of perceptive awareness in
a slightly different angle
and instead of asking myself what you
know
what does perceptive awareness is or
what self-consciousness is or what
self-awareness is
i i decided to ask the question
how would you really get in contact with
the world at most primitive
level how do our self-awareness
experience or perceptual awareness
or experience is more general um emerge
from the in in the first place right so
are they involved
with their lifespan and if you take that
perspective
right then you realize that actually
from the very beginning
uh we have that that traditional axis
a subject here relates to with an object
there is somehow
incorrect because in real life the way
we start actually perceiving the world
is with the subject here that relates
with another subject here in a very
proximal way
right so this means that we'll probably
need to shift the perspective in

instead of like taking like a tacit
visual spatial understanding of what the
perceptual awareness is
we need to actually expand it and be
more exclusive
and uh you know pay attention to the
fact that our first experience are
basically multi-sensory nature and they
also
proximal this in a sense like way
basically
before we meet other people's minds we
meet
other people's body and why is this
important is because there is a huge
literature
on uh you know what does it mean for
uh what what is a minimal self and
actually this is important for
not only for philosophy but also for um
for
science and robotics and um many many
other fields
because it is important to you know
correctly define what but really
what we really mean by a self and by a
minimal self
and there's a huge literature um
in from from various traditions that you
know um talk about minimal
self-consciousness minimal self
core self uh pre-reflective sense of
self
um etc and
i think if we switch the perspective and
take
on doors like a more inclusive
ecological and bottom-up

um approach and what the minimal self is
we avoid
a fallacy a center conceptual fallacy
so for instance i'm trying to depict
this fallacy with this picture here
so suppose you want to find out what
is a minimal
model of a self-organizing
system such as a plant or a human and
then if you
take if you endorse an adult centric
perspective then you might tempt it to
basically
take whatever structure we have as an
adult plant or an
adult human and squeeze those property
or try to you know extra extract the
elixir
know the structure the basic structure
of those property and then you say okay
so this is how
the minimal self model of that self
organizing system is right like you ski
squeeze the plant make it smaller and
say
this is what the minimum model is but in
reality we know very well that actually
this is not how
it looks right so uh for instance if you
can see here
that a plant at the minimal level you
know the the form the stroke
structure here is very different that
the ones that you have later on
in let's say the other life so this
means that we need to
if we want to extract like you know the
basic the skeleton of a certain you know

um system then we need a from
an ecological and dynamic perspective we
need to take into account
all these intermediary forms in order to
make sure that actually understand
what the plant is and what a human is
right
so this is a paper that we
as i said we recently uh published this
year so we decided to
really really go back to um the
beginning and to
question more carefully this idea of
like um
um combo treatment and homeostasis
uh and i want to start with the
disclaimer because this is very much
work in progress
and there is a lot of work to do because
this is as i said is
slow it's a shift in perspective so
there is a lot of work to do actually to
shake a bit you know the
the you know the entrance point to the
problem
and uh this is not an active inference
account
of co-embodiment and co-homeostasis yet
this is work in progress i'm doing this
on different levels
from you know um metabolism to
you know um more high level level
um like self-awareness
but because i need to start somewhere we
decided with my collaborators basically
to use the active inference approach
some of the insights of the active
inference approach to bring the

fundamental state of combat
and the co-homeostasis under the
spotlight because this is something that
we need to do
because it's like a snap mainstream so
we decided to okay so let's take a sort
of like meta
level approach use the active inference
lens and show that if you
if you um like the active inference
approach if you endorse it then you we
you need to take carefully into
consideration this kind of like very low
level
co-embodiment approach from the from the
very beginning
so this is what we're going to do in
this paper
so i'm going to start very briefly with
the background and
predictive processing framework i'm sure
that you are all
uh familiar with this uh so i'm not
going to spend
a lot of time explaining it but i just
have
two slides on it just because i need to
make sure that if someone is looking you
know
watching this and doesn't know what
active influence or predictive
processing is
we are on the same page and then i'm
going to spend actually more time
and trying to develop the um
this and motivate this need to actually
take a
bottom-up developmental take on what

perceptual awareness and experiences in
the first place

and define the notion of embodiment you
know stages of stays in common stasis

homeostasis but again i'm going to

define them through

this um bottom-up developmental lens

and i'm going to end with some um hand

waving

work in progress uh and i'm very open to
suggestions

uh because as i said this is like very

much an open question

and we need there is a lot uh more work

to do okay

so um i'm i i believe if you are all

familiar with the predictive processing

framework so the basic idea is that our

brains is is not a

passive sponge that you know

automatically somehow you know

passively receives information somehow

he

you know gives the motor output but

rather we should conceive the brain as a

sort of like active

system that anticipates the next sensor

input

on the basis of prior inputs such as

that actually can

map the world in a more accurate way

this idea very nicely fits with some

previous idea that having been developed

in parallel by uh the embodied cognition

paradigm approaching cognitive science

where they basically

this author like verily colleagues

basically

outline this idea that cognition should
be understood

as the enactment of the world and the
mind on the basis of history a variety
of action that the being in the world
performs

and in the sense like we cannot really
understand cognition in isolation from
the body and we cannot understand the
body in isolation from the environment
and we cannot understand either the body
or you know the relation between the
mind body and the environment
without taking into account the history
of those like

uh connection from the in the first
place right because this is not static
phenomena but it's more like
dynamic unfolded for good time because
this is what life is right

so this means that our ongoing
perception and cognitive processes and
action are influenced by prior events
and prior experiences so this is
something that

you know um embodied cognition
approaches and

uh predictive processing approaches
somehow agree

and a different level of explanation but
the basic idea is that

so you cannot consider this phenomenon
in isolation from

past events right um so this means

so and another i think common ground is
this idea that

humans are biological agents that emerge
as like

proactive survival enabled prediction
machines which are strive to
reduce uncertainty within a highly
volatile environment so we basically try
to make sense
of the environment out there because you
know the world is bigger than our heads
and we try to find a way to actually
navigate this bigger world
which is highly unpredictable and
potentially potentially threatening
so one idea is that in order to keep
track of this survival
and reproduction relevant bodily and
wealthy information
our brain solves that problem of
uncertainty by
generating self and world models and
it does so by extracting statistical
patterns of information from its embody
and worldly interaction
and there is a mathematical story behind
is i'm i'm not a mathematician i'm a
philosopher some
you know um leave that to the
mathematician but
i think what i need as a philosopher
i need just this idea that
you know our ongoing perception
cognition and action and process through
the lines of prior self
and well-related information right so
this is what i need the only information
i need
the only thesis i need now if
this is true then this means that
uh it becomes crucial to look at how
this experiences an action perception

gets off the ground from the outset
right because we need that lens to be
accurate
right in a dynamic and ecological way
one basic basic and overlooked aspect of
our
um embodied and predictive processing
approaches in both philosophy and
cognitive neuroscience is that brains
and minds and human bodies
first develop within another human body
uh and this is important point because
crucially while not
all humans will have the experience of
being pregnant or carrying a baby
the experience of being caring and
growing within another person is
universal so this is something that all
humans are concerned with right so
and this this is something important as
well because
way before we meet other people's minds
we literally meet on to logically meet
other people's bodies and there's
something i developed um already into
2017 where i looked at empathy and basic
form of shared experience but the idea
is the same
it's like okay so if we really want to
understand the relation between
uh mind body um and how
um you know embodied creature relate
with other with others
and the world we really need to go back
to this basic idea that
yeah before we meet other people's minds
we literally meet other people's body
so there is a lot of work previous work

that show
that looked at the homeostatic and
bodily mechanism operating
at early stage of developing in infancy
uh so uh post a birth right
uh we decided to take uh
a step further and to focus on pregnancy
uh because these are uh this is a
universal case remember
i i i'd like to emphasize this
because typically you know when people
hear the word pregnancy say okay this is
a woman's stuff
uh because well i i mean certain certain
people certain person can get pregnant
that
other don't so but if you take
the perspective the other perspective
then you realize that actually all
humans
are concerned with this experience right
where two individual organisms
literally grasp and grip into each other
and um contrary to the common view that
the
fetus is passively contained and sort
sometimes just put us like
solipsistically trapped within the
solitude and the silence of the womb
uh we will present evidence showing that
actually
uh there's a lot of going on there
there is an active in bi-directional
co-regulation between the two living
bodies and to live in organism
um and we need to start with some basic
definitions so first of all we're going
to

talk a lot about embodiment but we need
to make clear what exactly we mean by
embodiment in the first place
uh there is a lot of work already in the
literature under
the relation between the body itself um
i highly recommend this
like um very recent uh review by Keenan
colleagues coming from
Yorknottofove they have a very rich um
overview of
you know the relation between the body
itself and self-awareness
uh but here i'm going to use um my
analytical
philosophy card and i'm going to define
embodiment
in its minimal form to refer to the
instantiation of body properties by a
given individual in our case a human
in other words old talk about embodiment
Candice
conceptually ontologically linked to
talk about embodied beings
and then and their bodily properties
now the key starting idea here is that
brain and minds as i said it's like if
you're taking the body cognition
approach and also more recently like the
predict process inactive inference
approach it's like
they all seem to agree that the brain
and minds don't emerge in a vacuum but
develop and evolve as a system dedicated
to controlling and
observing the organic survival and
reproduction so basically
if you take the classical mind-body

problem instead of collect conceiving
the body as a sort of like
a vehicle or car that you know is it's
there to you know support and fuel the
mind
actually you should consider that way
around namely that the mind has
evolved to actually subserve the needs
of a uh organism right uh
of a body um of your human
body so you basically shift the
perspective which one which is
actually which comes first which one is
actually more important
so crucially however if
so if we endorse this perspective and we
agree with this shift then we need to
consider the fact that human bodies
do not emerge in a vacuum either right
so human bodies basically
emerge and develop within another human
body
so this means that we need to adapt and
extend the notion of
audimen to reflect this fundamental body
within a body case
which again i'm going to stress again
and again this is universal right it's
not constant just
like a part of the population
so but what does it mean to say that the
two agents
are co-voted well
uh in biological terms this means that
uh as we'll see later very shortly
is that the regulatory of the two agent
states
and particularly the need to maintain

that by physiological stability despite
the fluctuating and unpredictable
environment
is actively negotiated between the two
organisms that
share for a given amount of time
typically nine months but could be
you know shorter a bit longer common and
bodily and environmental uh resources
right and i'm i need to give you an
example so what exactly i mean by going
embodiment
um because i think this is a key point
so i'm going to ask you to walk with me
into a thought experiment because i'm
philosopher so i'm going to present you
with the thought experiment
so you imagine you um you have a friend
and you uh you know you you
you have to perform a sequel show right
and with your friend and you have to
share this pair of trousers here
so your friend puts his his left leg in
the left part of the show of the
trousers and you put the
right leg on the right part of the
throttle uh throughout this
and you need to coordinate your walking
such as actually you can walk as fast
as you can to actually reach a target
right an apple stay
um now uh
suppose that your friends as hitting his
toe against
rock stumbles and falls and experienced
them by fear so what
is happening in his body in his organism
right

so um this means that maybe your friend
is going to untrain you
in full in which case you can
say that the two bodies i mean you the
friends are called fully or you shared a
state of like falling together
and you can also say that well because
it's kind of like
negative events your friends your friend
is kind of like
experiencing fear and you also
experiencing fear as well while falling
right
so you share that state of um you know
fear not crucially however that
this is not necessarily so so suppose
for instance that
you know your friend is uh is a
master yoga teacher right uh and
while you are following you your friend
is somehow
you know managed to keep the balance and
you know
or the you know the tissue of the
trousers kind of like
you know destroyed and you know one is
standing at other one is falling right
it's not necessary so you can imagine in
case that actually this is happening
and also imagine a case in which let's
say well
why your friend is falling and
experiencing fear you find the entire
situation actually pretty funny
and instead of experiencing fear you
experience amusement right
so the two emotional states are not
necessarily shared so your friend is

experiencing fear or
your um experience like say amusement
right
so that's one example of like what we
can understand by
sharing a state now let us go back to
the case of pregnancy right
not that the case of pregnancy is very
different
because if you take exactly the same
example uh and
say that the mother is like some stumps
under
on a penis something and you know false
then there is no way in which defeaters
can say wait a minute
actually i'm not i don't like the
situation i don't want to fall i don't
want to
experience fear and i don't want to
experience negative you know
emotion i'm going to sit you know stay
put here you go ahead for
right so this means that actually there
is a way in which the two states are the
two organisms
necessarily such as the body itself
regulatory processes of one organism
necessarily
is linked to the um the other one
so it is important to distinguish the
case of pregnancy from the case of two
bodies like contingently happen and to
share a state
right and we can just define the notion
of like co-embodiment and saying that
two or more if you have twins or
uh more more than three to two

uh kids are co-avoided in this strong case if they are bound to share bodily and environmental resources for a set time period typically nine months for survival

purposes uh note that pregnancy is not only a case where we can find instances of this type of like stroke or

strong embodiment like foreign siblings but we decided to focus on the case of pregnancy because as a status is universal case all humans have you know experienced this so

there is already work on the fascinating phenomenology of shared embodiment and i highly recommend to to read it in what follows basically i'm going to very quickly

uh go to start with a very basic level of

actually what is happening at the you know from single cell going up to the organism and that kind of like shared

um shared embodiment again this is fascinating a huge literature when you think about it we start with like this tiny little round thing with another person's body and then we you know become ourselves

um so uh again i'm going to be very quick and very sketchy on this like fascinating pro

you know process but i'll i'll try to flag you know the

the crucial steps and the crucial stages right uh so the first stage is the process of implantation so

basically when
there is a dialogue between the
fertilized egg and
the mother body right and this is
actually this is the crucial step of a
pregnancy
a successful pregnancy um because for
instance
most of the miscarriages happen during
this step
um this is a like a sort of like
negotiation process going on between the
two
organisms saying uh and like any
negotiation
things can go okay or
can go wrong right so this means that
both the blast to see the baby and the
endometrial tissues they need to
actively regulate gene expression
transcription factors signal pathways
inhibiting factors and growth factors
during that implantation process
this is highly complex process and very
fragile
and very delicate if something goes
wrong it can go to miscarriage
or um later on
and it pulls
at this stage the maternal body
literally self modifies some of
through differentiation into specific
type that support
implantation right so the key idea here
is that embryo and the mother's body
need to coordinate with this cells
within that
very tiny implantation window right

now if this process let's suppose is
like successful
then the coordination goes uh to the
next step
between week seven and thirteen more
movements um it involves
self-organization can start
so and um and the self-organization
um uh interestingly involves movement
which is actually very important because
especially for the active
inference community because as you say
as you can see
actually movement is present
very early on in our lives so
these movements are initiated from the
so-called
uh central pattern generator in the
early spinal cord which produces around
with eight very abrupt startles or the
entire fetal body
right importantly note that this
throttles which is somehow automatic
right as we think of like a sort of like
heartbeat right it's going you know
automatically
uh provide the minimal basis for the
most basic occasion of sensory feedback
because in doing so for instance
and might start to touch the face
and that's softly how you have my hand
you know autumn
automatically or unwilling to attach my
face and then i received this like
sensory feedback on my face from my
right hand or my leg might change
its position be flexed or extended or
both so this means that i might have

some minimal sensory feedback at the stage that that fools the possibility to respond and adapt to these movements this means that self-organization and fetal movement initiate as a biomechanical process that progressively builds sensory and regulatory capacities so for example as i said by being stimulated by startup the fetus perceives that the hand is movable and that touching the face rather than the uterine wall with hands uh might feel different because i'm more sensitive here on the face that actually i can i can feel my i can feel my fingertips so if i touch the uterine wall but if i touch my own face i have a double touch right okay so with this at hand we can go further to this notion of homeostasis and bear in mind that this is all happening so we have a developing body within another body now homeostasis typically refers to the organism's need to maintain unstable organic internal environment despite to fluctuate in external environment right and the homeostatic regulation typically entails using feedback control mechanisms to keep the senses viable of that internal environment within the ranges compatible for the survival in this case for compatible human

survival

now it is well established that this

control involves uh

automatic actions like motor reflexes

endocrine immuno immunological and

autonomic processes

um that are driven by the feedback

inputs and the resulting prediction

error

which is the discrepancy between the

expected bodily states

uh at a homeostatic set point

and its actual level as signaled by the

sensory inputs

from the body uh and as early as 1929

uh it has been you know proposed that uh

maybe the control

body homeostasis consists to the primary

aim of

a brain function right and this is

again this uh underlying exception is

that the fundamental drive towards like

self-preservation to take care of this

body this self

is essentially linked with constraints

of acting within the body's visual

states right

so it has been classically proposed to

define the homeostatic set points at the

expectation the means

of prior belief about the states the

body should inhibit

inhabit uh and or homeostatic range as

the variants of this

are prior beliefs now

going back to our case right so let's go

back to the to you know just for one or

the primitive level

so many years it has been assumed that the fetus has no metabolism of his own right uh now is nowadays is well established that actually the placental and the fetal liver working on them as a sort of coordinating multi-organ system to provide unnecessary nutrients serving the fetal metabolism and growth and this is important because you know if you talk about homeostasis we need to talk about metabolite because glucose is the most important fuel for fetal metabolism and ultimately for homeostasis so this means that maintaining an optimal balance as a continuous supplies of nutrients from the mother to the fetus is critical from the development and from the fetus standpoint is basically critical for maintaining his own um you know homeostatic balance now note that the fetus may be able to control and save some energy to some extent through many complex and dynamic processes but importantly however it is oversupply and under supplied of nutrients by the mother right uh which are process controlled by the pregnant person that can permanently uh program the fetal metabolism adversely right so this means that um um the the way the mother um and handles her own um metabolic processes significantly

impacts the way
you know uh the babies um
you know has his uh self-regulatory
process maintained
and there is accumulating evidence that
there is a direct link between mother's
adverse bodily and mental health states
like depression and anxiety during
pregnancy
and child medium mental and physical
development but this is
this is not only for this is valid not
only for the negative states but also
more like
neutral states such as food preferences
and flavors
that are passed on uh from the mother to
the infant so there is a strong strong
codependence there
now the the
key into the into the intermediary
conclusion at the stage is that the
emerging self-organizing
organism uh human organism is not only
self-centered but also
crucially co-embodied and co-depending
upon homeostatic processes within the
environment right so that's the basic um
thesis up now now
we need to take this a step further
because
we need to talk about also action
anticipation and not just like
our prediction or perception
so we saw up to now that homeostatic
regulation
requires a model that enables inference
perception and prediction action

selection importantly however
the brain initiates regulatory responses
prior to homeostatic alteration
provided it can be anticipated so this
means that you
basically act to go to the fridge to
seek food
uh because you are hungry but actually
you aren't
you're not you you're not uh wait
you're not waiting for the starvation to
happen
to actually go to the fridge you do it
well before that actually is going to
happen so you anticipate to that state
so this means that higher level that
enable prospective control are leveraged
with
two essential components like inference
and prediction
which brings us to allostasis which is
defined as anticipatory homeostatic control
basically you take action into the world
because you need to regulate your inner
homeostatic
you know balance but you're doing this
by taking action into the world
and importantly you're doing this by um
anticipating
you know that state you don't really
expect to be on the edge of like
you know dying from starvation to
actually take that action
if you are optimal you know um
surviving um organism so this means that
the process whereby
the agent selects action that will most
probably bring about the desired sensory

entries

while explicitly or implicitly modifying
the true causal structure of the
environment

switch to the guaranteed recurrence of
those inputs over multiple
time steps um so this means that i can
act differently in different
environments

i can go uh to a fridge if i'm hungry
uh but if i'm in the desert i could
you know if i expect to see a fridge uh
and you know i don't see it uh then i'm
going to probably die instead of looking
for a fridge i should probably look for
i don't know banana tree or coconut tree
right so the environment defines what
type of expectations you should have in
a particular case

now there are two points that are
important here

note that uh for the individual living
organisms homeostatic

i mean we can conceptually distinguish
homeostatic processes from allostatic
processes but in real life
those go hand in hand uh

hand in hand right to the other two
faces of the same coin

this is because we uh humans we are not
static isolated creatures we are dynamic
beings depending on the rich
environmental resources

and we depend upon social interaction to
achieve survival and reproduction
and to give it like the basic example
that i really like

is that like the breathing like um it's

it's a perception so it's an
open-ended process right we are
constantly exchanging information
with environment via breathing right
uh and this is a dynamic process because
my body moves
right my thoracic cage is moving all the
time
and this means that those type of
processes like um allostatic pro
allostatic processes and homeostatic
processes even though we can
conceptually distinguish it in reality
they are actually coca
this means that perception in action
homeostatic and allostatic
self-regulation are coupled
strongly coupled in a dynamic load that
keeps the organism permeable
to and connected with this surroundings
for better and for worse because when i
breathe in i can take
in the air right uh that is necessary
oxygen for my survivor but i can also
get in i
take in the coronavirus you know and you
know
eventually potentially die right so we
are open system for the better and for
us in that sense
right um the second point is that
uh going back to the our case here is in
pregnancy is that self-regulatory
architecture is multi-layered this is an
interesting
point here right it's like it's already
complicated for individual but imagine
for actually two individuals that are

literally gripped into each other right
because the baby's regulatory processes
are coupled with the mother's
homeostatic and anaesthetic
processes and this is achieved mainly
through the placenta
which is an fmo and intermediary or
that enables vital and biological
exchanges between the two bodies
for example it allows the infant to
breathe despite the lack of proper
proper lung regulation and ensure that
the infants is fed despite lacking
proper eating effectors
not again that placenta is the
relational organ excellence but is also
a universal organ
whereas typically people associate the
center with like the female body
and a certain type of population to say
to a certain type of
persons but we all
have use all humans have used
successfully
this organ in order to survive and you
know i don't know do
philosophy cognitive science and active
influence later on in their lives right
and this is important to keep in mind
now
if i'm right this means that a dynamic
and complex system such as the human
body needs to be able to play a double
game
in order to survive and potentially
reproduce it has to successfully
maintain
sensory states within a certain

physiological balance
but it also has to flexibly change the
state in order to adapt a constantly
changing environment
so if we look at the human body through
through this dynamic lens it becomes
obvious that
what happens in between the organisms
environment i mean the boundaries
between the two play a key role
in making sure this game is correctly
played
and flexible enough to actually maintain
the organ is alive
uh so this means that we need to define
the krishna uh critical notion of
boundary or in between
right we need a new conceptual toolbox
in order to make sure that we get that
process correctly
right and this is something i'm
currently working on with some
collaborators
and some some of you probably know that
you know the notion of
marco blanton uh marco blanc has been
uh you know advocate as potential
candidates to you know promising way of
conceptualizing this
key notion of the boundary the medium
between the two and again this is like
uh uh working progress and there is like
um a lot of work to be done on that side
but for
for this talk i i really need to retain
this like
very simple idea that homeostatic
and anticipatory self-regulation depends

not only on the body states
uh and but also on the external
environment
and more crucially on the relation
between the two
so we can just define co-homeostasis as
the bi-directional process of
co-regulation that happens in between
two combated organisms
and homeostasis involves self-regulation
through
others couplet and homeostatic and
aerostatic loopy
states not
that and this is something that we have
to you know um you know work a bit with
the reviewers on our paper and they were
kind of like right
pressing clarification on this point
because it's important to bear in mind
that
if we talk about bi-directional process
doesn't mean there is a
asymmetry there or absence of conflict
on the country actually things can go
wrong
right um in the sense like
uh it can be
that you know one uh
one agent actually takes the balance too
much in one sense than other
and to give you an example like what
exactly what i mean by that
so think of a uh highway
right uh so you can have
that's that's a bi-directional highway
traveling from north to south
let's say that you have typically more

cars coming from north
to the south from the mother to the
infant
but you have if you can have one single
car traveling from south north
uh and that car is like you know i don't
know carrying bomb
then actually that's irrelevant right
it's not the quantity that matters that
actually the quality of the information
that is carried between the two
organisms actually have many many cases
and basically
during the pregnancy you know mothers
our pregnant person develop um diabetes
or other
um immuno
immune immunological disorders right uh
so that's not necessarily
uh a straightforward process as i said
and i'm emphasizing this this is a
constant negotiation between the two
organisms
it's a dynamic open-ended process that
can go okay
but can also get go wrong
now the big claim the radical claim we
want to make is that
given the necessary and universal
coinboardman sustaining the human body's
self-regulation and self-organizatory
processes from the outset
the early homeostatic regulation should
be fundamentally considered as
co-regulation
hence communist thesis now
uh in a further work we want to develop
this idea

of uh when exactly
and how the brain starts to build a self
and world model
in the womb right uh because it's like
it's a key question from uh for
the you know predictive processing
framework and active inference community
so
how exactly this is happening when
exactly this is happening
this is highly speculative um and very
much work in progress
um so it's it's an open question whether
the fetus is able to generate
self and oil models and if yes at what
stage of development
some people argue that this might be
possible only after the basic talmud
cortical connection are established
in the fetal brain around 20 weeks of
gestation this is because
so the the main reason is because
after this time point sensory
information can be transferred to the
fetal brain
uh before they establish of this uh
telomere cortical connection the fetus
just acts
on reflex automatic mode however
other people uh argue that through the
central part
generating seemingly chaotic
exploitative movements they also
um can generate a sort of like a sensory
motor coupling
and or some very basic primitive sensory
experiences
um again this is like an open uh

question

and this is something that i address um

in a preprint currently under review we

are two of my collaborator uh adam

safron and jonathan delafield

uh we look at how exactly self-awareness

no so how exactly the fact that our

first experiences

are co-embodied impact

may impact the way we understand current

theories

of conscious experiences in general

and what we our provocative take is that

you know our consciousness study focused

too much

on visual experiences uh and a visual

spatial way uh perspective endorsing a

tacit adult centric and visual spatial

perspective

and we want to say that perhaps

conscious experience are not all about

seeing red apples

to do you know famous example

so what are the implications of

switching the perspective you know from

this like what we understand by minimum

self and minimal self models you know

from

endorsing more ecological bottom up

perspective

rather than adult centric and um

vision-based perspective again this is

very much work in progress

something that we are working uh on

right now

but you know just like bullet points

uh if we switch to perspective then we

need to question the individually static

and solipsistic approach of
understanding of the perceptual link
between the ancient environment
we also need to question the
adult-centric thesis of sort of like
fully-fledged
emerging individuals like you know
athena
greek god or this athena coming out from
zeus head
so rather than humans emerge within
other humans and develop through close
interaction with other humans
uh we also need to question the idea of
a self-awareness for minimal self
developing independently of other cells
this is because as i hopefully you know
uh convinced you that we basically were
ontologically depending from the others
from the very beginning
at the very basic biological level and
then the question we need to also
question the idea of a perception being
a solitary and distant deva
like seeing a red apple uh but rather we
need to
you know see a perception as an active
interaction
active engagement with environment like
for instance touching human being
or coupling the interceptive like
signaling like the breathing so the way
i feel the air inside my body the where
i
take the air from outside um and also
important that
perception of that perception you know
that perception comes before the seeing

the red
um apple and if this
approach is right this means that that
early primitive perception
influences the way we perceive and see
the red apple
later on in life and to take home
messages that uh
when it comes to understanding nature of
our perceptual experience in general i
also have an argument for this in this
paper but um
you know we are trying to build a case
for this idea as well
and conscious experience as well um the
infant is father to the men
uh to the human sorry uh to paraphrase
this
um metaphor by
the english poet waltwood and
this is it i would like to thank my
collaborators for working with me on
this like
radical project uh i would like to thank
the
uh organizing of the active inference
lab for inviting me
uh i also i would like to ask you that
if you are interested to collaborate
with me
along those lines of like paradigm shift
please drop me an email because i'm um
very open
to these ideas and to develop further
um and yeah this is it and thank you for
your attention
thanks ana that was really interesting
maybe you can unshare

and we'll have a little bit of time to
discuss
cool so if anybody wants to post a
question they're free
to do so i have a bunch of thoughts in
question but
one piece that right at the beginning
you mentioned that
the historical nature of inaction and it
just shows that sometimes when people
are talking about inactive
approaches they're oh you know picking
up this pen it's the snapshot
inactivist but actually the implications
of action
are that we have to think about history
and development so i really appreciate
the way that you framed it in
in these terms um so
maybe just to get things started where
does active inference come into play
here
that's pretty big question but where
does active reference come into play
yeah so i think active influence come
into play in
the way we actually understand how the
minimal
self is constrained right so how the
minimal self organizing system
is somehow negotiate that exchange of
information between you know
his own self model and the environment
right
so the radical idea we want to put
forward there is that the
closest environment and the most
primitive environment that we have in

the first place
is another human body is not some sort
of like
objective environment but another
subjective i mean lived up lived
you know uh body right uh so i think
that's
kind of like radical idea and this means
that we basically we need to not
add an extra level even more dynamic to
the active inference to say
it's just not enough to basically just
act
and to make sense of that action in the
world and the movement but actually
we need to take it as a sort of like
double way it's a core coordination
uh between the two organisms right
so it's not just like function
it's just like co-action and core
movement
uh so this means that it's even more
messier than we thought at the
very beginning it's like if you take you
know the
the standard uh approach right so you
have the environment the agent and then
the agent needs to build the model of
that environment basically by
contrasting you know whatever the
self internal self mother says and
whatever extracting the mapping of the
environment has
right now you need to take into account
that that's a kind of like environment
it's not only
just like and a another agent out there
that you need to interact with

and that agent can be i don't know
another human being or a robot right
but actually you share common
resources computational resources with
that another agent right
and you need to actively negotiate those
resources
on the fly right even though there is
kind of like a um
asymmetry because obviously the pregnant
person has way more power
uh you know she can uh he can move
around
and um you know act in the world
but uh that co-coordination happens from
from the very beginning
that makes me wonder how is it related
to niche construction that's something
that
axel constant has of course worked a lot
on
so how do we think about niche
construction is there a niche a shared
niche is one the niche of the other
yeah so that's that's i mean this is
like as i said this is an emerging field
and i have
something like a zillion questions an
interesting question to us
and one of those is like related to the
niche
constructions for instance of this like
homebody
so the the pregnant person right so
there is like
very nice work showing that uh because
also the mother is embedded in a wider
you know social and physical environment

so we have cultural
right constraints on that pregnant
person and there is work showing that
you know for instance
like um you know the the the maternal
language
for instance influences you know the the
way
the infant will later on cry and if and
also if the
niche so to speak for instance is
bilingual because i don't know the
father speaks one language and the
mother speaks another language and also
impacts the way
for instance the the infant later on
cries or you know coordinates right so
this but again this is emerging evidence
we need
way more work to actually carefully look
at
that exchanges between you know uh the
embodied subject the corn body subject
the pregnant subject
and it's like close environment but what
we know for sure
is that clearly i mean um
i mean the the the the social background
the cultural background and
the physical background is really really
important right has direct direct
impact right and actually there are
there are
studies that show i can't remember the
on the top of my head
the name of the authors but i yeah they
show quite nicely that actually
you know uh uh i don't know french

babies or german babies
they don't cry the same way because the
language spoken by them the mothers
has different intonation right uh
so the auditory input that you can have
uh in the room actually
significantly impacts right so this
means that if you change
niche the cultural nature change
environment you're also
going to change the way the baby later
on will
somehow express you know this regulatory
process like crying right i'm not happy
right that makes me think about how even
as
adults we're still bounded by externally
constructed membranes we rely
on a broader society to bring us glucose
and to take our waste
and so there's a critical period just
like you had with the seed
absolutely a critical period where just
like for any other complex system
it's going to have bifurcations and a
lot of leverage points early in its
development when it's literally
more fluid but later on these things are
different in kind of
uh detail but not in type so extremely
yeah and i think it's it's really
important because we
we tend to we need to i mean i i forgot
for that slide but i think
it's important to bear in mind what we
call the invisible basis of the
aids ice bank right so certain things we
really take for granted so

if you go again to the plant metaphor
right
so um or the tree metaphor so think of a
tree right
so you walk on a street so let's suppose
you're a martian
right and you land on the earth and you
see a tree and understand how the tree
works what is a tree
right so if you limit yourself to
only to what you see above the surface
of the earth right
like the trunk the leaves you know um
then you might miss some important bits
of information of how exactly tree works
because the tree also has
roots and basically the way is negotiate
the nutrients
through that basic roots invisible ones
actually
you know significantly impacts how later
on i don't know the trunk or
the leaves are constructed right so we
really need to this is our motivation to
take
to endorse this like bottom-up
developmental and ecological dynamical
perspective because
we really need to go back to the roots
invisible one
that we take for granted because as we
said
rightly point out maybe there is
something going on at that critical
stage that if you know
act upon that critical stage then you
avoid later on bigger problems
right that you can't modify um so you

need to you know act
exactly at the time t for a very
specific reason
but also those roots are highly
connected with
i don't know the nutrients in the earth
in this case like
i know mother's body which the mother
body is kind of like
connected again with the closest
environment i mean the way he is like
supported or the cultural background
that the type of support she gets from
her close environment
later on and then you know i don't know
going higher up
and it's like policies that you know
governments can have
and you know climate change and other
things right
so it's everything is somehow a very
open and
uh precarious system and i think this is
i mean the way i understand and this is
why i'm interested in this like
entire you know new active inference
approaches because well as the name
shows is like it's an active thing right
it's like it's constantly dynamically
moving then we need to make sense of it
right in a very um you know
rigorous way but still you know um it's
it's an open-ended process
right yep and for the tree visually the
roots are there
but with humans our roots are often
transient or they're cultural so
it also asks us to move beyond a visual

representation of being tangled in a web
and you can't move but rather to see
these dynamical processes
that um go through the developing embryo
so here's a question
from the chat um anna asks
saying that they just came to see your
talk after hearing about it just a few
minutes before it began so that's so fun
that we can connect with people who are
just hearing about these ideas maybe for
the first time
and the question is um i would like to
ask for background references
to start studying this theme so what
would be
some key references or some key
questions or topics to look up for
somebody who's kind of wanting to
follow you on that freeway on that
research trajectory that you're on
so um so there are several
so it depends on the background right so
uh uh there is
there is very nice work on the phenology
side
so in philosophy right
so there is a nice work by um
alaina by and also
by honeydew and um
i think i i i hope i'm not misspelling
her name
but if you google hannibal she has a
recently last year paper with
i'm sorry i'm really bad with names um
so that's a very good reference starting
point
uh i also wrote on this uh so if you

google my name

i started to work on this already in
2017

i i worked on a basic form of
self-awareness

right um and also

there is nice work on um

um the the the touch and effective touch
component

right by uh maria laura philippe

and laura cuccinelli they have a very
nice review paper and a special issue
that

i edited it is called such a special
issue

called the relationship basic forms of
self awareness and they have a paper
there and actually there are quite nice
papers there as well on this topic

uh but more overall i wanted to say that
you know i have

um the twitter handle uh public

and there is um i also have my website

and uh the email address so i'm very
happy to take

uh emails and i can basically afford

um a normal um you know complete
reference because on the top of my head
i'm pretty sure i left aside

important references and i don't want to
you know say i i

you know i said this references i forgot
those references i want to you know keep
diplomatically so email me and i'm going
to send you

references what's interesting uh

and it reminds me of the way that now
the internet

is allowing us to be in very tight
feedback loops
more like a co-embodiment of knowledge
you're able to share your knowledge and
your perspective
on things and then somebody can ask a
question and now we're in a very
tight proximal communications wise cycle
where we can actually uh update each
other on oh it was actually this 2017
citation she was talking about
not the other one but if that was only
uh once in a while conversation we
couldn't actually develop
at this rate so that's quite interesting
again as adults who are co-constructing
each other's
ecological niche um
okay one thought i had and if anyone can
write more questions
is in most insects so as an entomologist
i
was thinking of insects most insects lay
eggs
and you had said that um the placenta
and the co-embodiment of pregnancy is
universal
and maybe it's mammalian in fact it's
defined
in humans oh yes yeah yeah universal to
humans yeah
so the egg for the insect contains
everything it needs
to get to the relevant life stage like
to hatch from the egg and then be
whatever
stage of development it's in um except
for the ants

well there's other um exceptions here
but in the ants
the larva is um after the egg it hatches
into a larva very quickly
and then the larva is actually in a
two-directional relationship and the
larva are called the social gut
and they have a one-ended gut and so
they're being fed
food and actually digesting it and in a
two-directional hormonal
relationship and so it was so
interesting that it's almost like
the length of the dialogue that the
the length of the intergenerational
transfer
is it one day is it one year is it 18
years
is it permanent are we always
co-creating
the length of that handoff and the
richness of the handoff seems to have
something to do with
the evolution of complexity so i was
wondering what you thought about that
yeah so this is amazing i mean i i mean
when you look at nature it's like
mind-blowing
i mean i i i was about to say it's like
we i mean
i'm talking from a philosophy
perspective i can safely say that you
know
philosophy has been kind of like
adult-centric
and you know uh you know human-centered
anthropocentric right um uh but if you
want to

really understand this like how you know
mind and body works
and you look at the the nature more
general then you realize
wow this is really fascinating and so
um you know we can learn so much so
you you you are telling me about the
ants but i i recently read something
because i started to be fascinated by
this idea of like placenta
right and i've read that for instance
some lizards
right so if you have exactly the same
species of blizzard
right and some of them lay eggs
and some of them actually have babies
right
exactly the same lizard and you ask
yourself why and then you realize that
actually it depends
on the climate where they are
if they are warm climate or a cold
climate right so if they are in warm
climate they lay eggs because the sun is
kind of like you know
but if you are in a cold climate then
you're going to keep the egg inside
right
and make the baby protect the baby
inside right and then
only after that baby kind of they safely
put it into the world right
so that's a very nice example of as you
said it's like the
the the flexibility and adaptability
that we can have to actually know the
make life happen and carry on right
because in this case you basically have

exactly the same body
um but the way you carry
on life is depending on exactly that in
between interaction that you have with
the
environment right in a cold environment
in one case
and warm environment another another and
that impacts the way you basically call
you know the way you you know um
self-organize and reproduce
it is really fascinating i really hope
that basically
you know um animal scientists and you
know
human brain scientists would work hand
in hand because i'm pretty sure will
somehow unlock many of the so-called
mystery of the
human mind and human body just by
looking carefully what is going on to
our
you know uh friends and neighbors right
animals right
it's an excellent point and in animal
cognition especially
it's phrased as like the development of
the cognitive capacity
as if that was the end point you know
you go from zero to 60
and then you're at 60 and then you
cognitive decline you know tap on the
brakes
but there's a richness and it's
different cognitive agents
which actually brought me to this
question which was um
i was definitely with you that the adult

centric perspective
that you raised in your first anecdote
at the um
conference and it's oh it's all adults
in the room so let's go
with this that it would be fallacious to
have a
adults-only perspective and maybe it
might be also
somewhat of an error to have like a
child only philosophy just whatever the
child
appeals to them let's just go with it so
how do we structure
in the intergenerational conversation
and learn from each other
yeah so that's that's a that's a big
question
so i i think for what we really wanted
to do in this paper
was like something very very modest
namely to draw attention on the fact
hey guys pay attention we have this lens
yeah it's kind of like tacit let's clean
it a bit and expand it right
so i'm happy to basically expanded it
and take that kind of like bottom-up
perspective but obviously
i don't want to bring everything to the
to the root so to speak right would be
like
exactly the same fallacy to like focus
just like like on the fruits like the
apple
in the tree and to focus on just on the
roots right i think what really needs is
an explanation of that dynamic process
right

how that unfolds how that emerged how is
connected with the wider environment
and with other minds other
bodies other animals um
i don't know if it's true i need to
check but at certain point somebody told
me
i think i saw it on twitter that
actually if you take a human body
and you take the cells and the human
body numerically
actually have more foreign cells than
actually human cells in your body
numerically because all you have was my
microbiota anything right
so if you if you take just like the
quantitatively
then we basically are more than another
than actually myself
which is like i i need to check that
information if that's true that's
actually quite interesting
one one thought on that um number of
cells is actually the co-embodiment
framework
moves us beyond a numbers game for
example
there could be a very big system and a
small system like the earth and the moon
and yet the moon is relevant for this or
that process on earth
and that's because they don't have to be
symmetrical if it was a
tug of war then it enforces a symmetry
which enforces a narrowness but actually
it
it doesn't really what if it was really
influential cells or less influential

cells

that can be accommodated in an active

inference um or in a

dialogue based framework but less so in

a sort of there's only one way to

collaborate like tit for tat

or game theory which is where a lot of

the models of pregnancy

uh just are still scratching their head

right um

here's a little of a meta science

question

if anyone else has any they can post to

is it was really cool

how you um demarcated

your thoughts on being a philosopher and

working with a formal theory

so what are your thoughts on working

with formalisms

as a philosopher or working on these

transdisciplinary teams

where there might be colleagues with

clinical psychological

math and a philosophy background so how

do we work together on these teams and

address these topics that are really

big yeah that's that's a that's a good

question and

sometimes it's like twitter is quite

good for that because it's like he just

wants me to get this like the

this conflict in the nutshell so i

remember somebody tweeted at some point

you know when there's all this

computational

approach is kind of like emerging uh and

uh regarding like the clinical

population they were saying

bad news for clinicians they would have
to
learn mathematics and statistics right
to build computational models right
and which somebody replied i said
bad news for mathematicians and
statisticians they will have to go to
hospitals and talk to the patients right
uh so there is i i like that
kind of exchange because this is the
problem that we have in the sense like
we really need
a dialogue between you know the world
out there you need to go
there talk to the patient gather the
data observe the insects you know
observe the human right you know um
spend countless you know
um hours is like measuring how many eggs
you know what they're laying and
everything uh and in the same time you
also need something more rigorous that
will allow you to basically you know
build models that you know you know
instead of like
how should i put this is like as if you
you you you test
hypothesis instead of a testimony in
real life you you let mathematics
actually take the risk of that
hypothesis and if it's strong that's
fine because it's not going to
die it's just a wrong hypothesis but in
real life if you test hypothesis
and you're wrong then actually you
you may die right so i think that's
should be a very nice dialogue between
the two uh two approaches

um should be seen like a sort of like
dig in the same tunnel but from the toe
opposites
sides and hopefully meeting in in the
middle

uh personally i like so i'm i like to
even though i have a philosophy
background i like to consider myself as
an interdisciplinary researchers
so i have collaborators from all
fields i collaborate with clinicians i
call

i'm doing experimental science so i'm
running studies

i'm collecting data i also collaborate
with people on the more formal side
so we're building kind of like you know
um

simulation and formal approaches to a
certain certain phenomena
especially connected with my main
interests which is

self-awareness minimal self-worth
alteration of minimal self-worth

um the relationship between the self and
the

body and co-embodiment and um so on and
so forth

but uh but yeah i i do and this is why
this is where i

said to the last in the last part of my
talk in the last slide that i'm very
open to collaboration

uh especially on that side because i
really need to i don't

really think we reached a point where we
basically need to have a dialogue
because it's like uh we have so much

information

and we realized that this

life out there the mind and the body and

everything is so rich

that we cannot we cannot solve it just

like we just like one single discipline

or one single tool we really need to

you know join forces and um a dialogue

it's going to be a bit more messier a

bit more trickier

because we don't have the same

conceptual toolbox so we'll have to do a

lot of

work beforehand so to clarify

what exactly we mean by that right in

different fields

and i think this is where philosophy can

be useful because we kind of like

obsessed with this idea

what do you mean by you know give me a

definition to make sure that actually

we're

we're not all over the map you know

we're talking the same phenomena

about the same phenomena but also i

think it would be dangerous for

philosophy to

to somehow develop this conceptual

toolbox completely cut off from

whatever you know you know the the

things out there the empirical world or

you know the mathematical models

actually showed us right so we need to

keep those gates open and um you know

exchange and dialogue and collaborate

wow thanks for that awesome response

um to call back to your beginning slides

you had the armchair philosopher and

that's a term that people know
just like uh in america we say monday
morning quarterback
just like looking back at the past and
um the armchair philosopher is not so
different right and that made me think
of i think therefore i am
and the idea that by actually getting
into some sort of altered state of
inference
by meditating or by doing some other um
thought
ritual it would be possible to reach
insight through thought
and the way that you just framed the
answer was like
hypothesis testing is an action that we
take in the world
it's not just a thought and when we
frame hypothesis testing as
action about the systems that we care
about like the next generations
then it makes us think about working on
teams
and on accountability and on
reproducible science
all these things that are sort of like
on the side
of the armchair philosopher and the
armchair philosopher might also care
about
participation in science but that would
be a secondary conclusion
whereas if hypothesis testing involves
stakeholders
and it's also uh something that's
happening in the loop it's a it's a
different situation so it's just really

like interesting i hope for those who
are listening to
hear what you're saying uh and actually
i keep
i i like to keep reminding my students
that you know i i put a picture of the
card
because i'm french and he's the his
friend just like the counter
anyway but i could have been aristotle
but i i don't know it's like those kind
of like classic philosophy we need to
keep in mind that
these people are i mean aristotle was
like biologists
right he was like constantly dissecting
things and you know testing things just
like
was like looking at what was going on
with the organism so aristotle and he's
considered to be
a great philosopher and similarly with
uh
with the god i mean he was like a
mathematician
right and he wrote a treaty on optics
right so it's like a vision and
everything else so
so um science and philosophy were kind
of like working hand in hand and i still
think they're actually working they
factor like i
really think that actually scientists
they
doing without realizing actually uh
philosophy
and uh some good philosophers actually
they're keeping an eye

or whatever is like scientists are
discovering whatever nature is telling
us right
um because it's like this is a constant
information processing writer we never
get to the bottom rock of the
reality right things are changing right
uh
it's like and i think what is really
fascinating with this
um example of the virus right let us
let's take the virus because this is
kind of like very recent and very
rarely invented and very timely right
now so if you are a classical
armchair philosopher then you have a
huge ontological puzzle
with virus because it's like this is
where it's keeping keep constantly
mutating right
and function of whoever is hosting
him right uh or you know whoever the
host of that virus is
creates different type of mutation right
uh and this is like it's an ontological
puzzle for
uh philosophies like what is the virus
which is like coronavirus which one is
like it's like
and you you feel like saying was
actually none of them and all of them
right because
somehow all that kind of like share this
type of information but it's
changing in function of whichever body
of number of bodies and that depends on
i don't know the culture and the food
they're ingesting or

i don't know different genetical
components and so many other factors
and i i i can literally say that we can
literally see
that actually this is like we literally
have
new beings coming you know new new
entities arising like
mutants mutant forms of like life
it's like constantly arising so quickly
right that
we can't even you know wait a minute let
me
figure out what is going on right uh so
i think we have a very nice
and humble example example of how
reality out there constantly is coming
to us
with like constant flow of like emerging
information emerging creatures
living being self-organizing systems and
sometimes it's like well we can't seem
that but
actually they kind of like dangerous
they can kill us
uh with all our science you know
um and that's i think that's a very
important point
we need to bear in mind yeah thanks for
that it
it made me think that it's not just
emerging
entities it's actually emerging
interfaces and
even if the number of entities were set
the interfaces would still be
combinatorial it would be explosions
just like words

even if the word vocabulary is set you
still have interfaces that can emerge
and then um as we say also here in the
states
putting descartes before the horse
absolutely an error
um but then it's actually kind of
something that tells
two layers of truth because it's
philosophy and science that are working
together and how we think about things
and how we
operate things and what do we hear
right now other than that there's a
disconnection between technology
and society information flow governance
decision making
all these things you just you know spin
around and pick a number
and it will be related to something like
that and so
that suggests that there is a
requirement for a new kind of a
conversation
yes i fully agree
do you have any other thoughts otherwise
this was really an awesome guest stream
and we all look forward to hearing from
you in the future but yeah you're always
welcome
you can come on whether you're the
author or not
yeah let's hope that you know uh this
will kind of like develop more
uh systematically and uh when i when the
situation
allows to basically you know
uh organize a platform or perhaps uh

a workshop in person meeting such as we
can actually
you know put this idea to work because i
think that um uh
just kind of like uh proximal
face-to-face interactions
have a different can can have a
different
type of outcome that you know the ones
we can have right now uh so we can
basically
you know consolidate the collaboration
dialogue on um
on a more uh uh you know concrete and
pragmatic
basis uh what i really i realized that
during the pandemic which was really
really fabulous is that
well uh as what's happening right now it
never occurred to me that basically well
well wait a minute we can have like a
zoom conversation with somebody from
california talking about this so just
kind of like
um you know crisis um
you know brought up the surface new way
of connecting
right and make us um richer in a way
but i also think that it's important to
have that you know range of problems in
place
in the first place to keep the balance
right right uh
because we have evolved going back to
the human nature and human organisms we
have evolved to basically move and
interact
if we don't move we don't interact we

simply die

and uh we need to you know keep moving

and interacting with each other and

uh i think that's that's also um

valid for different disciplines that's

what we basically need to

you know adapt and interact with each

other

as disciplines

as active lab in online participatory

lab

we will co-embody that event in person

when your region

is available and when the personnel are

there

then we'll scaffold it in a new way

because previously it was

all in person there was never seen a

need to live stream or never seen

a need to document to translate to make

it accessible

to make it available and we'll be able

to organize in a new way

and be able to help each other in ways

that we wouldn't have expected before so

i look forward to it

yeah me too great times

see you again bye bye-bye

[Music]

nice awesome stream thanks anna